MIL-C-22931/15A
20 January 1972
SUPERSEDING
MIL-C-22931/15
10 February 1968

MILITARY SPECIFICATION SHEET

CABLES, RADIO FREQUENCY, SEMIRIGID, COAXIAL, SEMI-AIR-DIELECTRIC, 2.850 AND 3.125 INCHES OUTSIDE DIAMETER, 50 OHMS

The complete requirements for procuring the cables described herein shall consist of this document and the latest issue of Specification MIL-C-22931.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

REQUIREMENTS:

Operating temperature range: -55 to +80° C.

Inner conductor: See table I.

Dielectrics:
Cable core and insulating layer: See table I.

Outer conductor: See table I.

Jacket: See table I.

Voltage withstand: See table I.

Insulation defects, spark: See table I.

Attenuation: See table I.

Velocity: See table I.

Capacitance: See table I.

Impedance: 50 ±1 ohms.

Voltage standing wave ratio: See table I.

Mandrel size for cold bend, bending, and temperature cycling tests: See table I.

Minimum bending radius: See table I.

Part number: M22931/15- (dash number from table I).

FSC 6145

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TABLE I. Dash numbers and characteristics.

Requirements 1/	Dash number						
	001	002	003	004	005	006	007
Inner conductor	Copper wire or tube		Copper tube, corrugated			Copper tube	
Overall diameter (±0.003(.08))	1.157(29.39) ±.004 (.10)		1.140(28.96) ±.005(.13)		do.	1.145(29.08) ±.005(.13)	
Dielectrics							
Cable core	Polystyrene		Polyethylene			Polyethylene	
Insulating layer	Polystyrene	do.	Not applicable	do.		Not applicable	Do.
Outer conductor	Aluminum tubing, smooth, seamless		Copper tube, corrugated		Aluminum tube, corrugated	Copper tube, corrugated	
Inside diameter (nominal)	2.850(72.39)		2.500(63.50)		2.460(62.48)	2.564(65.92)	
Outside diameter	3.125(79.38)		2.850(72.39) ±.015(.38)		2.850(72.39) ±.015(.38)	3.000(76.2) ±.015(.38)	
Jacket	Polyethylene	Not applicable		Polyethylene		Polyethylene	Not applicable
Diameter	3.275(83.19) +.030(.76) -.018(.46)		do.	3.020(76.71) ±.055(1.40)	do.	3.150(80.10) ±.065(1.65)	
Minimum wall thickness	0.070(1.78)			0.065(1.65)		0.075(1.91)	
Insulation defects, spark	Applicable	Not applicable		Applicable		Applicable	Not applicable
Voltage withstand							
Volts rms at 60 Hz	10,000 min		9,000		9,000	9,000	
Volts dc	18,000		16,000		16,000	16,000	
Attenuation (dB/100 ft max)							
30 MHz	0.07		0.084		0.093	0.084	
400 MHz	0.33		0.33		0.366	0.33	
1,000 MHz	0.55					0.62	
Velocity (percent)	92		93.3			94	
Capacitance (pF/ft)	22	do.	21.7	do.		21	Do.
VSWR					do.		
Initial (max) 500-1,000 MHz	1.12		1.15			1.15	
After bending and temperature cycling (max) 500-1,000	1.15		1.20			1.20	
Mandrel size radius	70(1778)		30(762.0)			30(762.0)	
Minimum bending radius	50(1270)		30(762.0)			30(762.0)	

1/ Dimensions are in inches; millimeters are in parentheses. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

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NOTES:

1. Cross reference information is as follows:

Part number	RG number	Former classification Class	Former classification Type
M22931/15-001	RG-234/U	N	I
M22931/15-002	RG-242/U		
M22931/15-003	RG-321/U	None	None
M22931/15-004	RG-322/U		
M22931/15-005	None		
M22931/15-006	None	None	None
M22931/15-007	None	None	None

2. Figures are for information.

Custodians:
Army - EL
Navy - EC
Air Force - 11

Review activities:
Army - EL, MU, MI
Navy - SH, EC
Air Force - 11, 17, 80
DSA - ES, IS

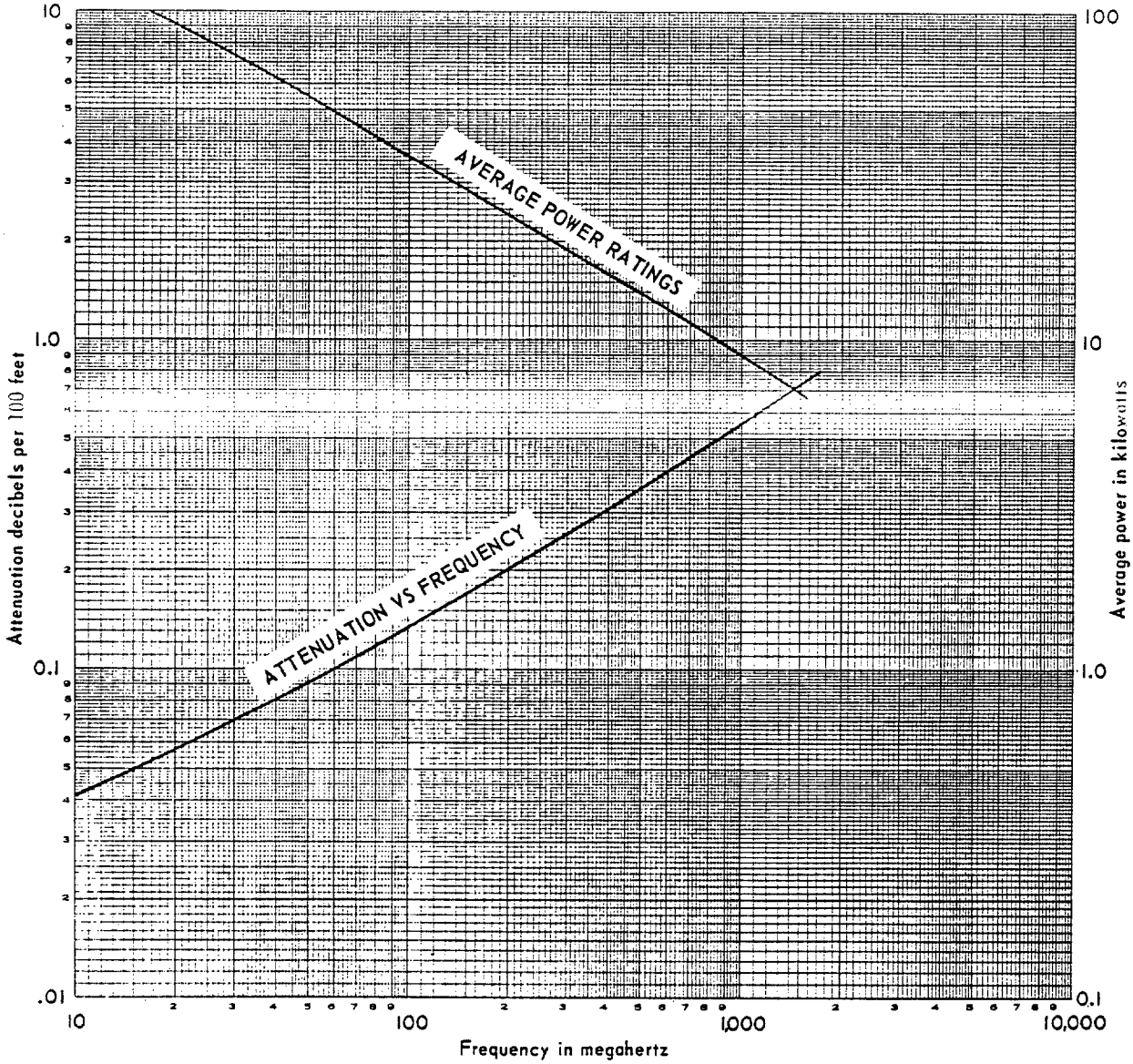
User activities:
Army - ME, AT, SL
Navy - AS, OS, MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 6145-0605)

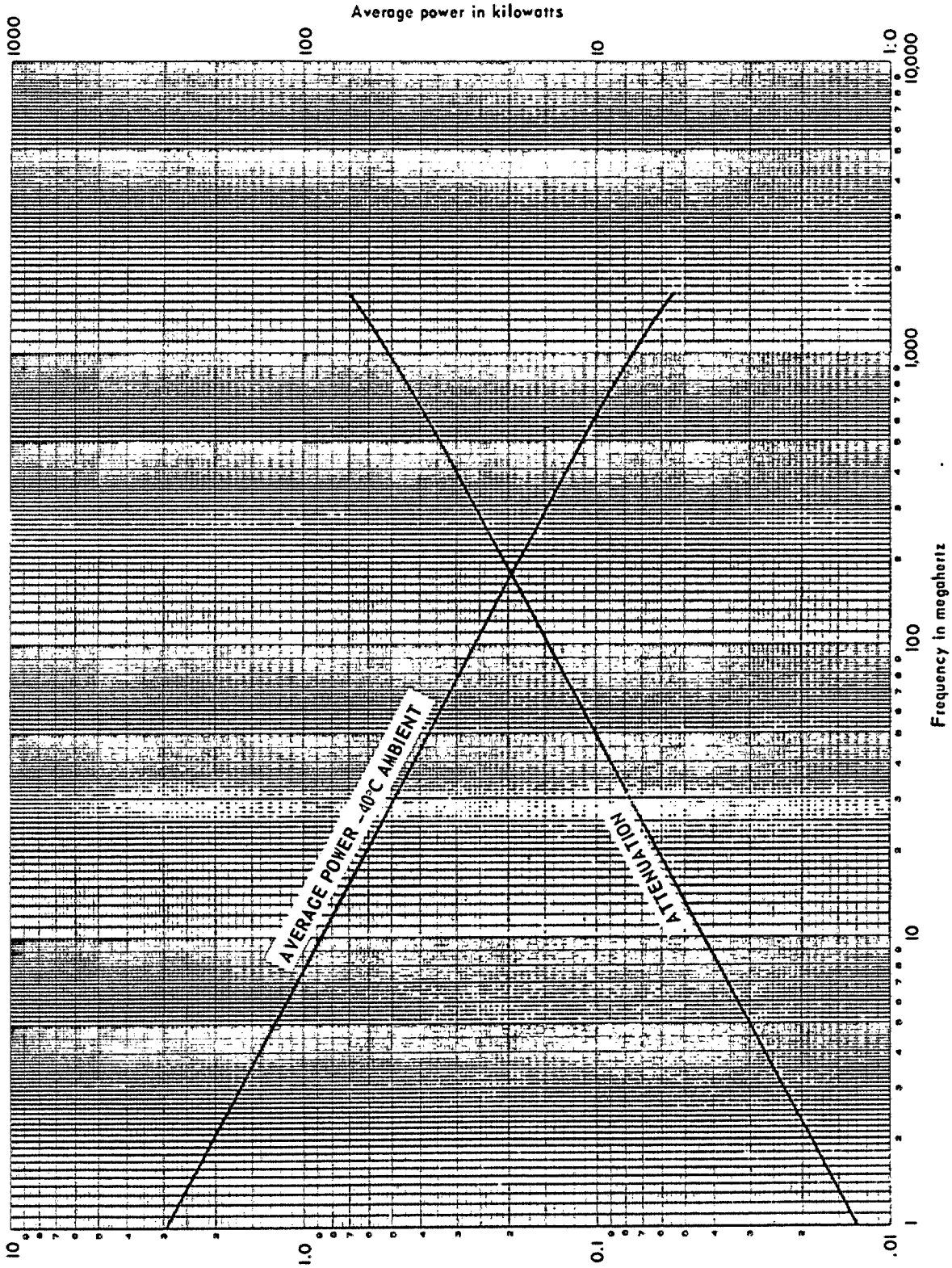
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 1. Attenuation and power rating curves for -001 and -002.

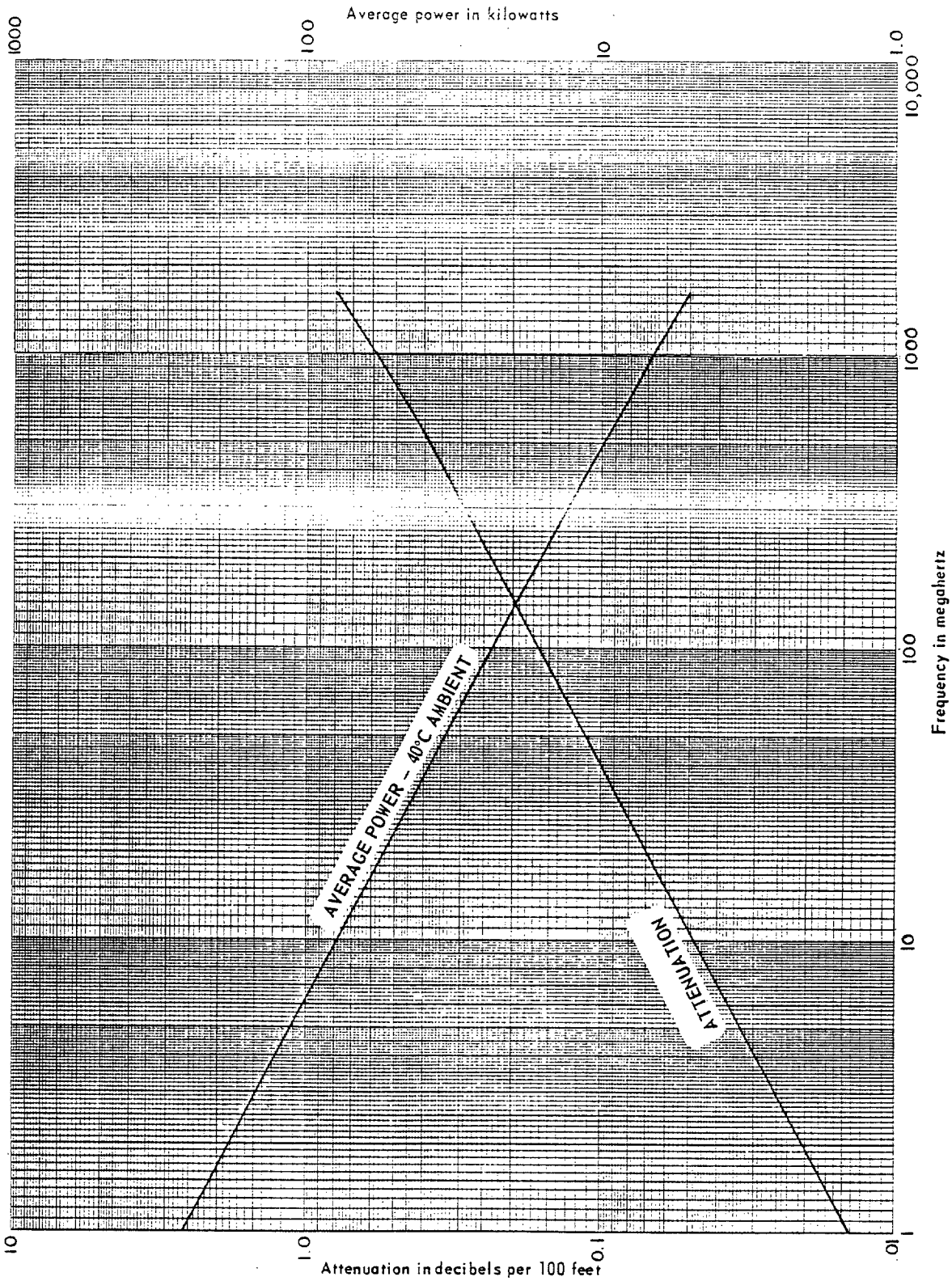
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 2. Attenuation and power rating curves for -003 and -004.

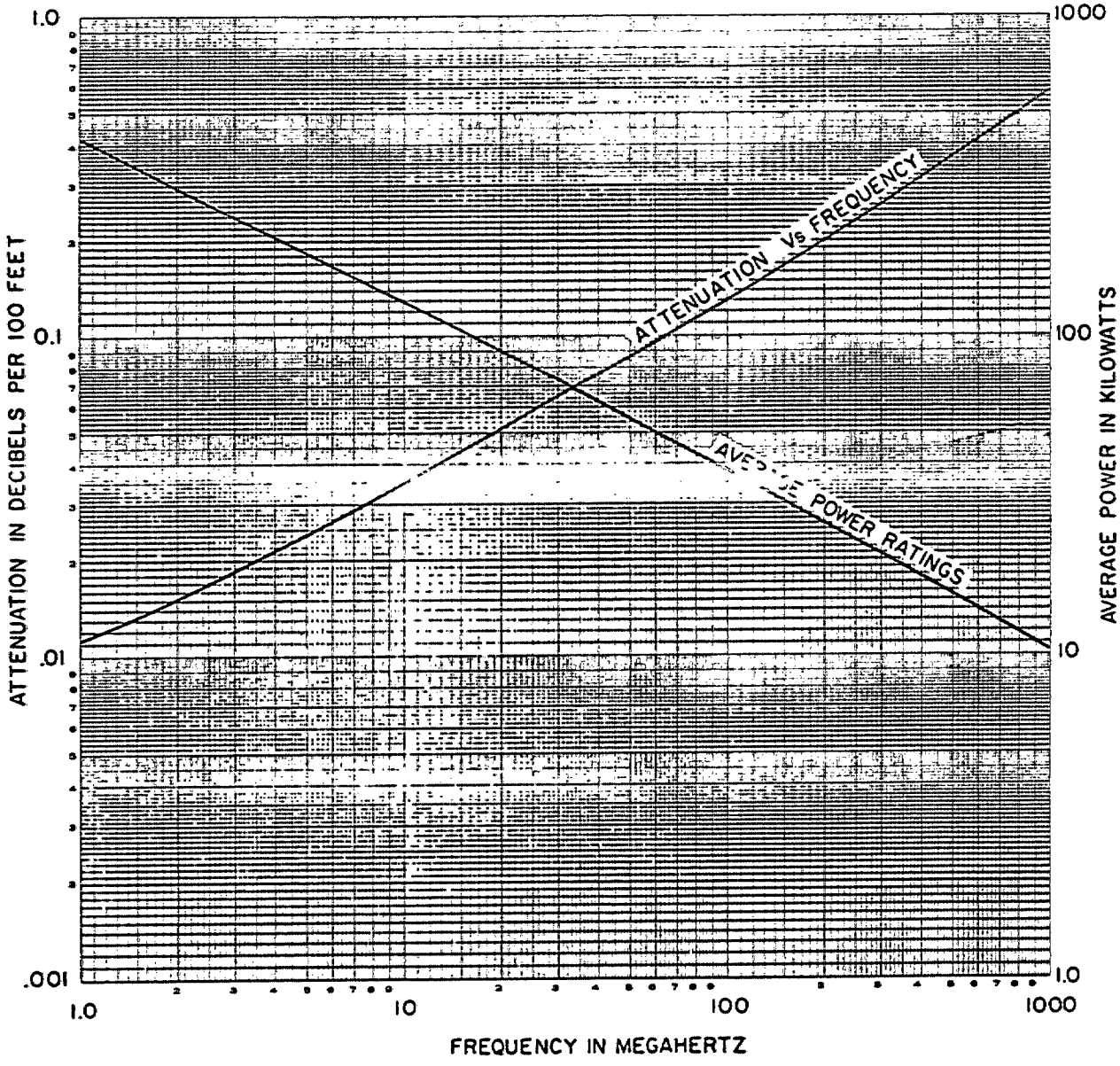
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Power rating based upon a 40°C ambient with a 45°C rise in center conductor temperature.

FIGURE 3. Attenuation and power rating curves for –005.

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POWER RATINGS BASED ON:
VSWR 1.0
AMBIENT TEMPERATURE 40°C
INNER CONDUCTOR TEMPERATURE 80° C

FIGURE 4. Attenuation and power rating curves for -006 and -007.

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